SUMATRIPTAN

Avoid in heart disease

For acute attacks only

CHEMICAL NAME 1-[3-(2-dimethylaminoethyl)-1H-indol-5-yl]-N-methylmethanesulfonamide (as succinate)

CLASS Antimigraine drug — selective 5-HT1B/1D receptor agonist (triptan)

TRADE NAMES Imigran · Imitrex · Suminat

THERAPEUTIC USES / INDICATIONS

- Acute treatment of migraine with or without aura.
- Acute cluster headache (SC injection, nasal).

ADMINISTRATION

- **PO:** at onset of headache phase (not during aura).
- **SC** autoinjector into thigh or arm.
- **Nasal spray:** one spray into one nostril.
- Not for prevention of migraine.

DOSAGES

- **PO:** 50–100 mg; repeat once after ≥2 h if headache returns; max 300 mg/24 h.
- **SC:** 6 mg; may repeat after 1 h; max 12 mg/24 h.
- **Nasal:** 10–20 mg; max 40 mg/24 h.
- No second dose if the first gave no response.

PHARMACOKINETICS

- **Onset:** SC 10–15 min; nasal 15 min; PO 30 min. **Peak:** PO 2 h.
- Oral bioavailability ~14%; metabolised by MAO-A.
- Excreted in urine. **t½:** 2 h.

PHARMACODYNAMICS

- Agonist at **5-HT1B** (cranial vessel constriction) and **5-HT1D** receptors (inhibits release of inflammatory neuropeptides from trigeminal nerve).
- Relieves headache, nausea, photophobia.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Tingling, heat, heaviness or pressure in chest, throat or limbs.
- Dizziness, flushing, fatigue.
- Injection-site reactions; taste disturbance (nasal).

ADVERSE EFFECTS (SERIOUS)

- Coronary vasospasm, myocardial ischaemia, arrhythmias.
- Stroke.
- Serotonin syndrome with SSRIs/SNRIs.
- Medication-overuse headache with frequent use.

CONTRAINDICATIONS & CAUTIONS

- Ischaemic heart disease, previous MI, coronary vasospasm, peripheral vascular disease.
- Previous stroke or TIA; uncontrolled hypertension.
- Hemiplegic or basilar migraine.
- Use within 24 h of ergotamine; with or within 2 weeks of MAOIs.

KEY INTERACTIONS

- **Ergotamine:** additive vasospasm.
- **MAOIs:** ↑ sumatriptan levels.
- **SSRIs, SNRIs, lithium:** serotonin syndrome.

NURSING MANAGEMENT

- Check cardiovascular history and BP before first dose.
- Give at onset of headache.
- Assess headache severity, nausea and light sensitivity before and after.
- Report chest pain lasting or severe.
- Monitor frequency of use (overuse headache).

PATIENT EDUCATION

- Take at the start of the headache, not during aura.
- Chest tightness may occur; seek help if severe or prolonged.
- Do not use more than 2 days a week regularly.
- Keep a headache diary.

KEY POINTS

- 5-HT1B/1D agonist for acute migraine.
- Contraindicated in coronary disease.
- Not for prevention.
- Never within 24 h of ergotamine.

SOURCES CHECKED **[68]** Imigran 20mg Nasal Spray (UK SmPC) **[69]** Sumatriptan 100mg film coated tablets (UK SmPC) **[70]** Sumatriptan 3 mg/0.5 ml solution for injection in pre-filled pen (UK SmPC) **[71]** Sumatriptan, oral (US label) **[72]** sumatriptan, SC (US label)

Full references p. 432–457